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NanoGene Technologies, Inc.

It was Friday, November 9, 2002, and Will Tompkins was both excited and concerned. The 41-year-old Biochemistry Ph.D. had quit his job at Eastern Institute of Technology's Advanced Materials Sciences Lab (AMSL) six months earlier to become CEO of NanoGene Technologies, a life sciences start-up based on nanotechnology. Over the previous six months, he and his four co-founders from AMSL had made tremendous progress in developing the underlying science that would enable the company to attract venture capital funding. Within the past 24 hours, Tompkins had participated in three meetings about issues that might have a dramatic impact on the future success of the business. The first meeting had taken place the day before with Paige Miller, a 1995 Harvard Business School (HBS) graduate who had been doing some consulting for NanoGene, and whom Tompkins was trying to recruit to join the management team. The second had taken place that morning between Tompkins and his four co-founders. He had just finished the third with Susan Stone, a venture capitalist (VC) who Tompkins hoped would become a lead investor for NanoGene's Series A funding.

Tompkins's co-founders included Don Rupert, the head of AMSL, as well as three fellow scientists from the lab: Mark Masterson, Ravi Rhoota, and Gary Garfield. The five had met that morning to discuss negotiating a compensation package that would entice Miller to join NanoGene as its VP of Operations. Miller had considerable experience in the life sciences industry as VP of Operations at a successful biotech company. (See **Exhibit 1** for resume.) Tompkins and his team were eager to have her on board. However, as Tompkins and Miller began to negotiate her compensation package, they soon realized that they were very far apart in terms of their expectations. Tompkins summarized the situation:

> Everyone likes Paige, and we have grown to respect her abilities a great deal. But she said she needed to make $175,000 in salary and have 3% of the equity (post-Series A financing). This basically means she will be making almost twice as much in compensation as the founders and have nearly the same amount of equity. All of the founders have worked very hard to build a culture of equality—it is important to us—but it is hard to imagine that a non-founder should make so much more than the rest of us.

Miller explained her view:

> I have been consulting in the biotech industry since I left my job two years ago, and I have gotten four job offers from post-Series A companies that have all been between $160,000 and

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Senior Lecturer Michael J. Roberts and Professor Linda A. Cyr and prepared this case. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management. The identities of individuals and institutions in this case have been disguised.

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> $180,000 in salary and 2.75% to 3% of the equity. I feel like I know what my market value is, and as much as I like Will, the whole team, and the prospects for the business, I don't see why I should settle for less than I am worth.

Tompkins's co-founders were disappointed and angered when they heard about the negotiations, and they raised several questions:

> Does this mean that Paige is not buying into our vision for the company? Do we even need someone of this caliber yet? Why hire a VP of Operations when we have no operations? I understand why we need an office manager and a bookkeeper, but what exactly are we buying for this big price tag?

The discussion was at times quite heated, and Tompkins knew that the issue was far from resolved when he had to excuse himself to go meet with Susan Stone, a local venture capitalist. His meeting with Stone got off to a much better start; Tompkins was encouraged when Stone told him that she was nearly ready to put together a preliminary term sheet and run it by her partners. The emotional boost was short-lived as Stone began to voice several concerns. Tompkins recalled:

> She said she liked the team and the technology that we were developing, but that we had made some very unconventional choices that she was going to have a hard time selling to her partners. First, was the large founding team. She said they see very few teams with more than three founders. Second, was the fact that we decided to split the equity equally among ourselves. She walked me through the numbers on the assumption that we raise $10 million for roughly 60% of the equity, which translates into each of us being left with only 3%, which she thought was quite a low figure at this stage of financing.
>
> Moreover, the fact that I would have the same percentage ownership, as CEO, also troubled her. And she felt the same way about my salary. I told her that each of the founders planned to draw $120,000 in salary. She thought that was above market for the co-founders who would be senior scientists—she thought $95,000 was market for them. And for me, she thought $120,000 was too low.

When Stone left his office, Tompkins sat for a moment with his head spinning. While he wanted time to think about the issues Stone had raised, he felt he had to turn his attention back to several business issues that would not wait: "In anticipation of funding, we had begun to interview candidates, and Paige had been stressing to me the importance of defining a set of hiring practices and a compensation plan for employees. Similarly, we needed to discuss our company culture. As we began to hire people, it seemed like it was time to formalize our thinking on this topic. "

As Friday came to a close, it was clear that each of the three parties—Tompkins, Miller, and Stone—had a lot to think through in the coming days.

Developing an Idea and Building a Team

The AMSL was one of the top material science labs in the world, and was associated with Eastern Institute of Technology (EIT). The lab was an eclectic mix of 50 or so scientists, including faculty from the ranks of EIT, post-doctoral students, and staff scientists, all of whom pursued a broad array of material science projects. The lab was well known for having made important discoveries that underpinned advances in semi-conductor technologies in the 1980s and early 1990s, and had turned its attention to nanotechnology and its applications in the mid 1990s. Nanotechnology refers to a set of technologies that allow the fabrication of materials at the scale of 10 to 100 nanometers

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(a nanometer is a billionth of a meter). The first commercial applications of the technology were on the horizon, and the area had become a reasonably "hot" area for venture investing. Between 2001 and late 2002, six start-ups had spun out of AMSL in the Boston area.

All of the NanoGene founders had worked at AMSL, although their work was in different sections of the lab. The scientists were working on a set of technologies that they knew linked some potentially exciting commercial applications of nanotechnology with advances in genomics. Specifically, work on the human genome project had identified specific genes that were the underlying cause of several diseases, such as certain forms of cancer. Moreover, it was widely believed that continuing advances in genomics would soon unlock the biological underpinnings of many more diseases. While scientists expected that it would be many years before such technology could be used for *treating* diseases, the prospect of using the technology to *diagnose* diseases before they became active in the body was widely believed to offer great, and earlier, promise.

One technique that had been developed to accomplish this relied upon isolating the specific genes responsible for a disease, and then testing individual patients to see if they carried those genes. Scientists were developing approaches to this testing that required depositing genes on a sophisticated substrate material and "slicing and dicing" the genes to permit this identification.

The founders of NanoGene had developed techniques and compounds that—when used as the substrate material—caused the genes to "stretch out and straighten up," making it much easier to do the cutting and manipulation that was required. The company's business plan was to first perfect, and then begin to sell, the patented substrate chips already "lined" with the particular genetic code in question (e.g., the two genes responsible to colon cancer) so that diagnostic testing labs around the world could simply wash patient samples over a chip and read the results.

By the summer of 2001, the founders decided that the technology indeed had promise, and were nearly ready to file the provisional patents—on behalf of AMSL—that underpinned this technology. As they continued to "gel" as a team, and as they felt more confident in the technology's promise, the scientists felt that they needed to talk to Rupert, AMSL's Director. Tompkins recalled the situation:

> By the summer of 2001, we knew what we wanted to do. We began to talk to Don because we genuinely wanted his advice. Don was an expert in substrate surface tension at the molecular level, and we knew his knowledge could really help us.
>
> Don was excited about what we were doing and said he was interested in becoming a co-founder. We were thrilled; Don had served on many companies' scientific advisory boards in the past, but had never agreed to be a co-founder before. Don said that he wanted to stay at the lab, and the terms of his employment limited his working with us to only one day a week, but we were sure that his endorsement would help us get funded and that his contacts would also be pivotal to our success.

Actually Starting the Company

In November 2001, the team met for dinner at Rupert's house to discuss formally incorporating the company. They discussed a number of important issues including equity splits, salaries, funding strategies, and naming Tompkins CEO. One of the co-founders, Mark Masterson explained:

> Each of us kicked in $1,000 to incorporate the company and get things going. We decided to split the equity equally. We also decided that we would begin seeking $600,000 or so in angel financing.

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> The money-raising discussion led us to start talking about what kinds of salaries we would take upon leaving the lab. We thought $120,000 in salary was reasonable even though there was variance in what each of us was making at the lab. Our salaries ranged from $55,000 to $90,000 depending on discipline (e.g., physicists were paid more than biologists) and whether you were a post-doc or on staff or an academic senior scientist, but it was well known that industry salaries were 50% higher than the academic salaries. I think it would have been disruptive to try to slice and dice the equity and compensation according to a more complicated set of principles; it would have upset the team.
>
> We determined that Will would take the CEO role by universal acclamation. He just exuded natural leadership and he already had contacts with VCs and with a law firm and had already learned much more than any of us about what it took to start a company.

At the time of the company's founding:

- Tompkins was a scientist in the biophysics group earning $80,000.
- Mark Masterson was a senior scientist in biophysics group earning $90,000.
- Ravi Rhoota was a scientist in the biology group earning $65,000.
- Gary Garfield was a post-doctoral fellow in biology earning $55,000.

Through early 2002, the team continued to work on their plans for starting a business, and sought funding to get the enterprise off the ground. The first 12 to 18 months of the company's life would involve perfecting the technology and proving that it would work oat a commercial scale. In Tompkins' words:

> The basic principles have been worked out and demonstrated in the lab. But, we need to put several different pieces of technology together, and then prove that they will all work at commercial scale. The angel round will really just scratch the surface—we need about $10 million to get through this 18-month phase. But the angel round will buy us a little time and let us get organized to raise some serious money.

Raising the Angel Round

One of Tompkins's contacts was a former CEO of a biotech company who was an active angel investor. After hearing the team's plan, and with Rupert's strong backing, he agreed to pull a group together to invest $600,000 at a $2.25 million post-money valuation. The deal closed in June 2002. At the time the financing was negotiated, the founders decided they would make their own stock vest according to the following schedule: 20% immediately, 20% at the end of the first year, and then the remaining 60% at the rate of 2% per month. Tompkins and the team began the process of looking for lab space and buying some of the equipment they would need.

When Tompkins resigned from the lab in May 2002, the angel round was pretty well negotiated, and he began working seriously on raising a more substantial venture round, and on recruiting a "business person" to the team. It was shortly after this point that one of the VCs whom Tompkins had met referred him to Miller.

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Recruiting Paige Miller

Paige Miller had six years of operational experience in biotech. After graduating from UCLA with a B.S. in Business, she spent the next few years working at a tennis club in the San Francisco Bay area before attending Harvard Business School to focus on manufacturing. Miller described her path:

> I played tennis competitively in college and wanted to continue playing, but court time was expensive in San Francisco, so I made a deal with this club that I would trade an hour of court time for an hour of my working around the club. Before long, I was spending more and more time around the club and was working full-time doing a lot of their back-office operations. But it didn't seem like this was destined to lead anywhere that I wanted to go—I felt like I wanted to make more of a difference in the world.
>
> At the time, there was a fear that the United States was falling behind as a world-class manufacturer, and some people suggested that this was an arena that needed bright young people. I spent a year working for Intel in their Washington, D.C., office doing research on U.S. manufacturing competitiveness. A friend suggested I apply to HBS, and I got in. I spent the next two years taking as many manufacturing and operations courses as I could.

Miller graduated from HBS in 1995. When she began looking for a job, she soon realized that she wasn't going to get a job through traditional recruiting channels. She set off on her own search:

> I decided that only three things mattered to me: I wanted to be in manufacturing, in biotech, or in medical instruments, and in Boston. I got a list of all the biotech and medical devices firms in Boston and wrote cold-call letters to the CEOs describing my background and telling them that I wanted to work in manufacturing. The CEO of BioMolecular Technologies followed up with me and we created a job where I would be the sidekick to the VP of Manufacturing. BioMolecular was a pre-IPO life sciences instrumentation company with about 120 people, and over the next couple of years it had gone public, and I had the opportunity to work in every kind of job in manufacturing as well as doing stints in finance and marketing.

In 1997, Miller took over as director of manufacturing and was in that role when BioMolecular was acquired by International Biotech. Miller was the youngest senior executive in this international company and was responsible for integrating global operations into the current site. After having seen BioMolecular go public and seeing it through its acquisition, Miller decided that she was ready to move on to new challenges. In late 2001, a business-school friend encouraged Miller to leave her current position without having another job. In early 2002, she began working through her VC network to look for another job, and took on some consulting work as well. She recalled meeting Tompkins:

> By summer, Will was trying to raise VC money in earnest and he was getting feedback about not having any business experience on the team. I was introduced to Will by some of the VCs I was networking with. I recall his asking me in one of our interviews what I would expect to make if I joined the team. I had received a handful of offers over the past twelve months to be VP of Operations at post-Series A companies. The typical offer was in the $175,000 salary range with 2.5% to 3% of the equity, and I told them I would look for something consistent with that. The initial reaction from Will and the team was, "oh, that doesn't seem too bad."

Tompkins recalled his thinking:

> I knew my strength was the science, not operations. I wanted a generalist—someone who was very much a team player and who was willing to do whatever it took to get the job done.

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> Initially, it was clear what she would do and the areas in which she would contribute. I expected that she would set HR policies, work on getting access to appropriate facilities, manage vendor relationships, deal with the landlords, and do the bookkeeping. She would do all the little things that would let us focus on the science.

Miller offered to do some consulting projects for NanoGene until it had raised its Series A round.

Acquiring the Intellectual Property

By late July 2002, NanoGene seemed well on its way, but before it could attract serious investors, it had to lock up the required intellectual property (IP). The key patents included four patents in Mark Masterson's name (but owned by EIT) which covered most of the key technology. One additional patent was in Ravi Rhoota's name (also owned by EIT) but was, in Tompkin's words, "critical to making all the technology work." In August, after a few months of negotiations, EIT agreed to license the required patents to NanoGene in exchange for 400,000 shares (15%) of the company's equity as well as a 3% royalty on sales.

Raising Venture Capital

Tompkins had been working on raising VC financing in an informal way since early summer 2002, and once the IP was firmed up, he began more serious meetings: "For the first few meetings, all five of us would pile into Gary's van and head over to meet with the VCs. After a few of these sessions, I got the feedback that this was sending a bad signal to the VCs, so just Ravi and I started going to the meetings."

NanoGene was seeking $10 million in Series A financing. The team had decided this was sufficient to fund them for 18 months, and to get them to technical proof-of-concept and pilot production. For the first year or so, all of the work would be on the "science" side of the project—perfecting the technology. Then, it would shift to commercializing that technology. It was anticipated that the team would be at roughly 30 people by that point, including the founders.

In September 2002, Tompkins was meeting frequently with VCs, trying to interest them in investing in the company. By October, several VCs were performing serious due diligence on the company. And most of the VCs had issues with some aspects of the team and the decisions they had made. Tompkins explained what he heard from the VCs:

> The first thing that raised a red flag was the very large founding team. The VCs said that about two-thirds of the teams they fund are pure scientists, but there are generally only two or three of them and usually one of them has had experience as a Chief Scientific Officer of an established firm, so they have experience driving the commercialization of technology. Here were five guys, none of whom had ever worked in a company. Both the number of founders and the collective business inexperience was very rare, in their view.
>
> Another big cause for concern was the way we had decided to split the equity. Their view was that a market exists for the way equity should be divided, and equally is *not* it. According to the VCs, biotech CEOs usually have 7 to 10% equity post-Series A and founder-CEOs have upwards of 15%. Typically, CEOs will retain about 2.5 times as much equity as VP-level founders.

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Their concerns were similar with regard to our decision to take equal salaries. They thought $120,000 was low for a CEO and too high for everyone else. A typical early-stage biotech CEO makes about $250,000 and 'market' for a senior scientist is $95,000. They didn't expect that I would make anywhere near that much because the $250,000 figure is for a much more seasoned executive, but they thought I should certainly make more than the other founders.

Tompkins had analyzed the likely distribution of equity ownership, assuming the company was successful in raising $10 million for roughly 60% of the equity, as the VCs had indicted. The VCs had also conveyed the sense that the company would have to create an option pool of roughly 13% of the shares, which Tompkins also modeled. (See **Exhibit 2** for Tompkins's pro-forma analysis of the company following the Series A financing.) Still, he was not convinced that there was a problem; he described his reactions to the VCs' concerns:

I thought the equal distribution of equity was important for a number of reasons. First, I didn't really view myself as more important than the rest of these guys—we all started this together and success would depend upon all of us, and I was CEO because my skills most closely fit those requirements. There was no argument among us with the notion of equality.

Building an Organization

In anticipation of the funding coming together by the year's end, Masterson quit his job at the lab in June 2002, shortly after Tompkins resigned. Rhoota quit his job at the lab in October 2002. Thus, three of the founders were drawing a salary from the money raised in the angel round, although they had decided to take the relatively low amount of $750 per week. In Tompkin's words, "We figured we could get by on a small amount for living expenses until the funding closed." Garfield was planning on quitting as soon as the Series A round closed. In Masterson's words, "Gary had a bigger family to support and mortgage payments to make. We didn't think it was fair for him to squeak by while we were waiting to raise the money."

Tompkins had kept in touch with Miller. In September, he asked Miller to begin doing some consulting work and at the same time enter into more serious discussions about coming on board in the Vice President of Operations role.

Defining the Hiring Process

By early November, the funding seemed imminent, and the founders decided to begin the process of hiring the next layer of scientists. The planned organization was that each of the other three full-time founders (i.e., the ones other than Tompkins and Rupert) would comprise the company's senior scientists, and that each would report to Tompkins, as would Miller. Each senior scientist would have reporting to him a scientist and two research assistants (RAs). This team of 12 scientists was anticipated to be in place through the first six months or so, and would expand to 25 or so by the end of the year. Although it was not yet clear exactly how many of each type of scientist would be hired, Tompkins's best guess was, "By year end, we'll have to hire 16 RAs, eight scientists, and one more senior scientist."

As the company got closer to funding, Tompkins encouraged his cofounders to begin to recruit some of the scientists and RAs they would need as they began to build the company. Miller—who was continuing to do consulting work for the company—described the discussion:

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People talked about how they should go about hiring someone—I realized they hadn't done much hiring. So, the issue for me was, how do I help these guys develop a better sense of how the hiring process should work, what principles should drive it. Should everyone at the company interview each candidate? Will has already said he is not sure he adds much value to the process. He, and some of the others, voiced the view that the decision should be up to whomever the candidate would report to. I feel that we should all interview everyone, and that any one of us should be able to veto a candidate. In a small company at a critical stage, one bad apple can really have a huge negative impact. I stressed to Will the importance of our meeting with the team to lay out a vision of a more formal set of hiring processes and policies.

Designing a Compensation Policy

Another issue that needed to be addressed concerned the policies around compensation for the new hires. Miller had done some work developing a sense of the market compensation for RAs and scientists, and she had defined a standard salary and option grant for each position level. RAs would be paid $45,000 and would receive 40,000 options, and scientists would be paid $72,000 and would receive 60,000 options. As the company had begun to interview to make its first hires, several founders had begun to push back, arguing that they needed more flexibility to attract the right talent to the business. Miller shared her thoughts:

I believed that you should use as many objective measures as possible to figure out what kind of offer to make in terms of salary and equity and then put forth your best offer. Philosophically, I didn't believe in negotiating over these things—you clearly defined the job, asked yourself whether the person could do the job, and then just paid for the job. I just didn't believe that the potential ill-will created by side deals was worth the trouble. I told Will that I felt the same way about my offer. Make the best offer and if the person accepts, great; if not, so be it.

We did some work on a compensation grid, deciding that the scientists we hired would all be paid $72,000 and all get the same number of options. But after interviewing a few people, one scientist said he is a serious triathlete and he knows that this will mean he needs some extra time away from work, and he would like to take 90% of the "standard" in exchange for an extra 20 vacation days. Another potential hire said his wife makes a lot of money and he would rather have less salary and more options. I really feel that we need to stick to our policies, but some of the co-founders say that we will never get the most creative scientists out of the academic labs if we are not more flexible. I can tell they kind of bristle at what they perceive to be my "bureaucratic" leanings.

It is also clear to me that we cannot give this level of options to each employee we hire. Basically, the VCs have signaled that they will give us an option pool of 13% or so of the shares (i.e., 1,400,000 shares assuming 10,565,000 shares outstanding post the Series A round). If my 3% comes out of that, this leaves 10%, or a little over 1 million shares for everyone else. I think it is fair to basically reduce the size of option grants associated with each position for every month we are in business—each step of progress we make reduces the risk in the company. The plan is to give a new hire an option grant on day one, and then for the options to vest 25% at the end of year one, and then 25% each year thereafter, but on a monthly basis. We don't anticipate making a second, follow-on grant to anyone until the option pool is refreshed in the Series B round. The VCs have told us that—since they are getting preferred stock and the options are for common stock—the strike price on the options can be 10% of their share price, or $0.15 per share in the model Will has run.

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Company Culture

Finally, both Tompkins and Miller had agreed that it was important to spend some time up-front talking about what they wanted the culture of the company to be, and how they could make this a reality. Tompkins recalled: "A couple of the VCs I met with seemed to make a point of saying that they thought the academic lab culture was not the right one for a start-up company, and they wanted to be sure we had a more performance-oriented model in mind."

Miller amplified:

> I felt pretty strongly on a couple of issues; one was instilling a sense of frugality in the company and the other was building a culture based on setting very concrete goals and objectives. One of the huge negatives of the dot-com era is the sense that a big round of venture capital makes you rich, and you should spend lots of money on office space and fancy furniture and a cappuccino machine.
>
> On the issue of goal setting, my experience with academic scientists is that the concept of goals is an abstract one for them. They are often distracted by a notion of discovery; they pursue interesting findings even when it takes them off track. In business, it is important to be focused on meeting specific, measurable goals and objectives. I think it is important to weave that into the fabric of the culture.

What to Do?

Early that evening as Tompkins prepared to leave the office, he realized that he, Miller and the VCs would have a lot to think about over the weekend. As he gathered a briefcase full of material from his desk, he thought to himself:

> On one hand, I think Paige could help us a lot. And, I do believe her sense of the market value of her services. But, I also feel as though we've arrived at an equity distribution for ourselves that is fair and reasonable, and based on some solid principles. So, I'm not sure how two seemingly reasonable points of view can be in such conflict. I guess it's my job as CEO to figure it out . . . . Maybe I should be getting more money for this job after all.

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Exhibit 1 Paige Miller Resume

PAIGE MILLER

EXPERIENCE

December 2001–present Consulting

Clients include BioMolecular Technologies (former employer) and several biotech start-ups.

1995–November 2001 Biomolecular Technologies Cambridge, MA

Director, Manufacturing 1997-Novemeber 2000

- Senior Executive responsible for all manufacturing operations including production, procurement, asset management, sustaining engineering, and distribution for this $150 million a year site
- Directed the activities of 85 people through 5 managers
- Responsible for $8.5 million annual expense budget
- Managed Corporate Information Systems for two years
- Point person for this site for all global operations integration since acquisition
- Took company from pre-IPO to acquisition
- Received every award and form of merit compensation given
- Youngest senior executive in the company

Manufacturing Analyst/Internal Consultant 1995–1996

- Projects included: Manufacturing, Marketing, Finance, Corporate/Engineering, Customer Service, Facilities
- Highlights: Led Company-wide team in the creation and implementation of New Product Development process; Top Grossing Sales Person for 1996; Performed "turnaround" of stockroom operations

1993 Intel Corporation Washington, DC

Research Project Consultant

1989–1992 Bayside Tennis Center Sausalito, CA

Operations Manager: Grew company into one of the premier tennis centers on the West Coast.

EDUCATION

1993–1995 Harvard Business School Boston, MA

- Masters, Business Administration

1983–1989 University of California, Los Angeles Los Angeles, CA

- Bachelor of Science, Business Administration

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Exhibit 2 NanoGene Capitalization Table, Assuming Venture Capital Financing

	Founding / incorporation 12/01		Angel Round 6/02		IP Deal 8/02		Assumed Series A	
	# shares	%	# shares	%	# shares	%	# shares	%
Tompkins	333,000	20.00%	333,000	14.70%	333,000	12.50%	333,000	3.15%
Rupert	333,000	20.00%	333,000	14.70%	333,000	12.50%	333,000	3.15%
Masterson	333,000	20.00%	333,000	14.70%	333,000	12.50%	333,000	3.15%
Rhoota	333,000	20.00%	333,000	14.70%	333,000	12.50%	333,000	3.15%
Garfield	333,000	20.00%	333,000	14.70%	333,000	12.50%	333,000	3.15%
Total	**1,665,000**	**100.00%**						
Angel Investors			600,000	26.49%	600,000	22.51%	600,000	5.68%
Total			**2,265,000**	**100.00%**				
EIT					400,000	15.01%	400,000	3.79%
Total					**2,665,000**	**100.00%**		
Option Pool							1,400,000	13.25%
Total VC investors							6,500,000	61.52%
Total							**10,565,000**	**100.00%**

Note: Assumed financing is $10 million in Series A Preferred Stock at approximately $1.54 per share.

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